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Date

- 1.) Given three integers n, a, b return n^{th} magical no, since the ans may be very large $10^9 + 10^6$.

APPROACH:

- (1) we count no. of multiple of $a \leq x$
(x is a trial value).
i.e. x/a
- (2) same way we count multiple of $b \leq x$
i.e. x/b
- (3) by inclusion exclusion count the multiples of both a and b .

$$(A \cup B) = (A) + (B) - (A \cap B)$$

$$\text{count}(n) = \left\lceil \frac{n}{A} \right\rceil + \left\lceil \frac{n}{B} \right\rceil - \left\lceil \frac{n}{\text{lcm}(A, B)} \right\rceil$$

if $\text{count}(n) \geq n \rightarrow n$ might be answer
search minimum x such that $\text{count}(n) \geq n$

$$l = \min(a, b)$$

$$r = n^k \min(a, b)$$

$$ob = \text{lcm}(a, b)$$

while ($l < r$)

$$mid = l + (r - l) / 2;$$

if $(\text{count} < n)$ $l = mid + 1$

else $r = mid$

}

return $l \% MOD$

$$T.C: \log(n^k \min(a, b))$$